Project: LoanGazeAI - Credit Risk Classification Model

Intelligent Loan Approval Prediction Through ML Gaze

Business Problem:

Develop ML models that can help the company for determining whether an applicant is eligible for a loan or not.

1) Business problem understanding:

Dream Housing Finance company deals in all kinds of home loans. They have presence across all urban, semi-urban, and rural areas. Customer first applies for home loan and after that company validates the customer's eligibility for loan.

Company wants to automate the loan eligibility process (real time) based on customer details provided while filling online application form.

These details are: Gender, Marital status, Education, Number of dependents, Income, Loan amount, Credit History and others. To automate this process, they have provided a dataset to identify the customer segments that are eligible for loan amount so that they can specifically target these customers.

In [4]:
```
# import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

In [5]:
```
# Load Data
data  = pd.read_csv('E:\\AI-ML_SB\\5.Classification\\Capstone Project- Classific
data.head()
```

Out[5]:

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome
0	LP001002	Male	No	0	Graduate	No	584
1	LP001003	Male	Yes	1	Graduate	No	458
2	LP001005	Male	Yes	0	Graduate	Yes	300
3	LP001006	Male	Yes	0	Not Graduate	No	258
4	LP001008	Male	No	0	Graduate	No	600

2) Data Understanding

1- Data Exploration

In [8]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 614 entries, 0 to 613
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Loan_ID            614 non-null    object 
 1   Gender             601 non-null    object 
 2   Married            611 non-null    object 
 3   Dependents         599 non-null    object 
 4   Education          614 non-null    object 
 5   Self_Employed      582 non-null    object 
 6   ApplicantIncome    614 non-null    int64  
 7   CoapplicantIncome  614 non-null    float64
 8   LoanAmount         592 non-null    float64
 9   Loan_Amount_Term   600 non-null    float64
 10  Credit_History     564 non-null    float64
 11  Property_Area      614 non-null    object 
 12  Loan_Status        614 non-null    object 
dtypes: float64(4), int64(1), object(8)
memory usage: 62.5+ KB
```

In [9]:
```
data.columns
```

Out[9]:
```
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',
       'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
       'Loan_Amount_Term', 'Credit_History', 'Property_Area', 'Loan_Status'],
      dtype='object')
```

In [10]:
```
data['Loan_ID'].nunique()
```

Out[10]:
```
614
```

In [11]:
```
# Droped unimportant column as per feature selection

if 'Loan_ID' in data.columns:
    data.drop(columns=['Loan_ID'], inplace=True)
```

In [12]:
```
data.head()
```

Out[12]:

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	Coapplic
0	Male	No	0	Graduate	No	5849	
1	Male	Yes	1	Graduate	No	4583	
2	Male	Yes	0	Graduate	Yes	3000	
3	Male	Yes	0	Not Graduate	No	2583	
4	Male	No	0	Graduate	No	6000	

In [13]:
```
data['Gender'].unique()
```

Out[13]:
```
array(['Male', 'Female', nan], dtype=object)
```

In [14]:
```
data['Gender'].value_counts()
```

Out[14]:
```
Gender
Male      489
Female    112
Name: count, dtype: int64
```

In [15]:
```
data['Married'].unique()
```

Out[15]:
```
array(['No', 'Yes', nan], dtype=object)
```

In [16]:
```
data['Married'].value_counts()
```

Out[16]:
```
Married
Yes    398
No     213
Name: count, dtype: int64
```

In [17]:
```
data['Dependents'].unique()
```

Out[17]:
```
array(['0', '1', '2', '3+', nan], dtype=object)
```

In [18]:
```
data['Dependents'].value_counts()
```

Out[18]:
```
Dependents
0     345
1     102
2     101
3+     51
Name: count, dtype: int64
```

In [19]:
```
data['Education'].unique()
```

Out[19]:
```
array(['Graduate', 'Not Graduate'], dtype=object)
```

In [20]:
```
data['Education'].value_counts()
```

Out[20]:
```
Education
Graduate        480
Not Graduate    134
Name: count, dtype: int64
```

In [21]:
```
data['Self_Employed'].unique()
```

Out[21]:
```
array(['No', 'Yes', nan], dtype=object)
```

In [22]:
```
data['Self_Employed'].value_counts()
```

Out[22]:
```
Self_Employed
No     500
Yes     82
Name: count, dtype: int64
```

Create a new column as per requirement --> Income

In [24]:
```
data['Income'] = data['ApplicantIncome'] + data['CoapplicantIncome']
```

In [25]:
```
data.drop(columns=['ApplicantIncome', 'CoapplicantIncome'], inplace=True)
```

In [26]:
```
data['Income'].describe()      # Continous var
```

Out[26]:
```
count      614.000000
mean      7024.705081
std       6458.663872
min       1442.000000
25%       4166.000000
50%       5416.500000
75%       7521.750000
max      81000.000000
Name: Income, dtype: float64
```

In [27]:
```
data['LoanAmount'].describe()
```

Out[27]:
```
count    592.000000
mean     146.412162
std       85.587325
min        9.000000
25%      100.000000
50%      128.000000
75%      168.000000
max      700.000000
Name: LoanAmount, dtype: float64
```

In [28]:
```
data['Loan_Amount_Term'].unique()
```

Out[28]:
```
array([360., 120., 240.,  nan, 180.,  60., 300., 480.,  36.,  84.,  12.])
```

In [29]:
```
data['Loan_Amount_Term'].value_counts()
```

Out[29]:
```
Loan_Amount_Term
360.0    512
180.0     44
480.0     15
300.0     13
240.0      4
84.0       4
120.0      3
60.0       2
36.0       2
12.0       1
Name: count, dtype: int64
```

In [30]:
```
data['Credit_History'].unique()
```

Out[30]:
```
array([ 1.,  0., nan])
```

In [31]:
```
data['Credit_History'].value_counts()
```

Out[31]:
```
Credit_History
1.0    475
0.0     89
Name: count, dtype: int64
```

In [32]:
```
data['Property_Area'].unique()
```

Out[32]:
```
array(['Urban', 'Rural', 'Semiurban'], dtype=object)
```

In [33]:
```
data['Property_Area'].value_counts()
```

Out[33]:
```
Property_Area
Semiurban    233
Urban        202
Rural        179
Name: count, dtype: int64
```

In [34]:
```
data['Loan_Status'].unique()
```

Out[34]:
```
array(['Y', 'N'], dtype=object)
```

In [35]:
```
data['Loan_Status'].value_counts()
```

Out[35]:
```
Loan_Status
Y    422
N    192
Name: count, dtype: int64
```

In [36]:
```
continous = ['Income', 'LoanAmount']
descrete_categorical = [ 'Gender', 'Married', 'Education','Self_Employed', 'Cred
descrete_count = ['Dependents', 'Loan_Amount_Term' ]
```

EDA

For Continous Variable apply describe()

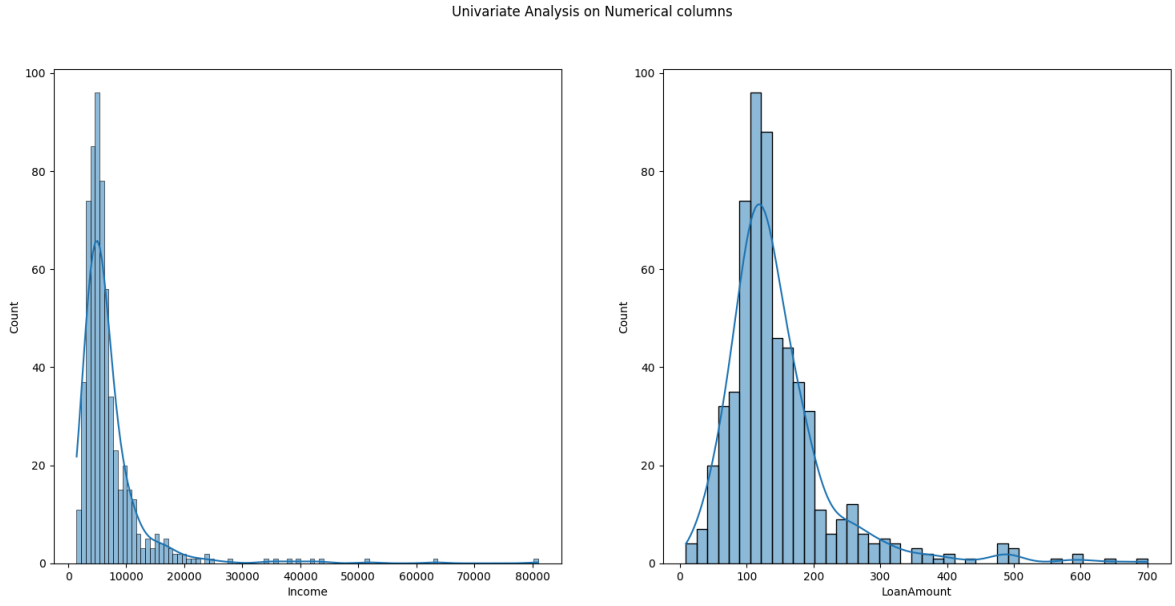
In [39]:
```
data[continous].describe()
```

Out[39]:

	Income	LoanAmount
count	614.000000	592.000000
mean	7024.705081	146.412162
std	6458.663872	85.587325
min	1442.000000	9.000000
25%	4166.000000	100.000000
50%	5416.500000	128.000000
75%	7521.750000	168.000000
max	81000.000000	700.000000

For Continous Variable apply Histogram

In [41]:
```
plt.rcParams['figure.figsize']=(18,8)

plt.subplot(1,2,1)
sns.histplot(data['Income'], kde=True)
plt.subplot(1,2,2)
sns.histplot(data['LoanAmount'], kde=True)

plt.suptitle('Univariate Analysis on Numerical columns ')
plt.show()
```

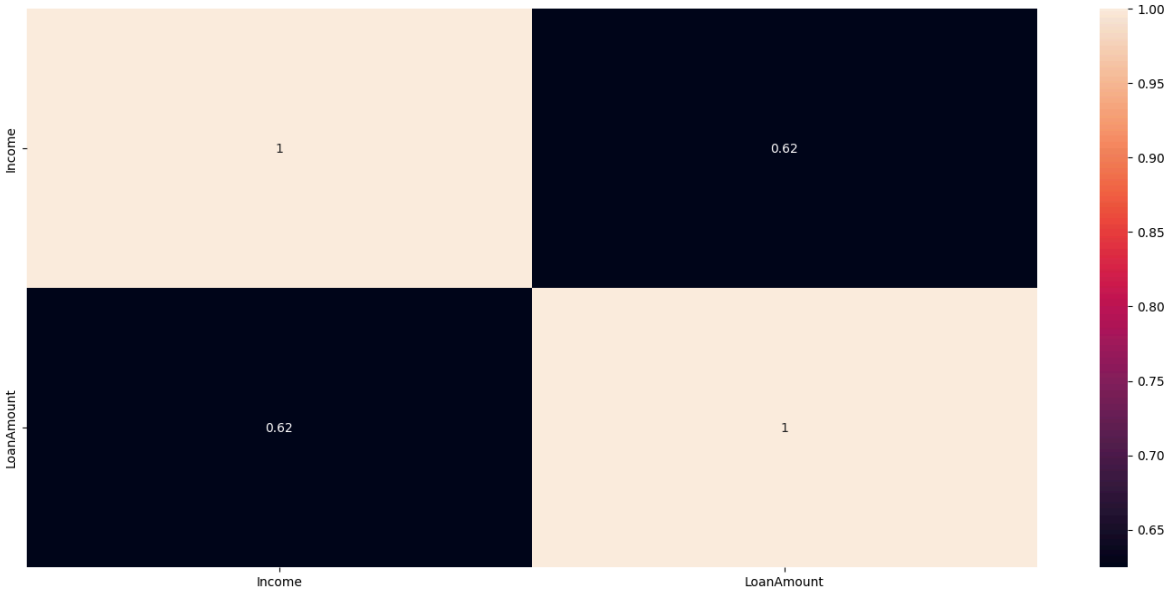


Both Income and LoanAmount are right skewed

--> very less no. of people having high salaries & very less are applying for High loan amount

For Continous Variable applied Correlation matrix (heatmap)

In [44]:
```
sns.heatmap(data[continous].corr(), annot=True)
plt.show()
```

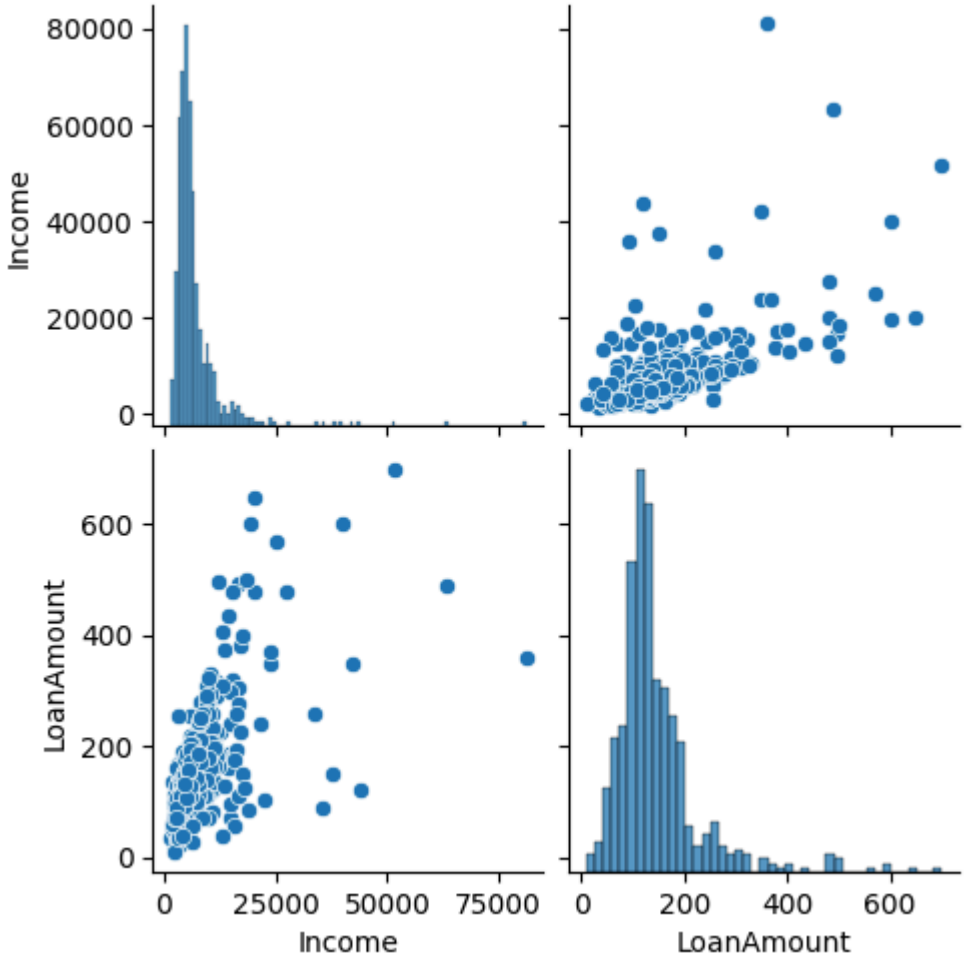


Income & LoanAmount are correlated with 62%

--> Higher the income, Higher the LoanAmount

For Continous Variable applied pairplot to observe scatter plot

In [47]:
```
sns.pairplot(data[continous])
plt.show()
```



Income & LoanAmount bothe are +ve correlated

--> as Income increases, LoanAmount also increases

For Descrete variable apply describe()

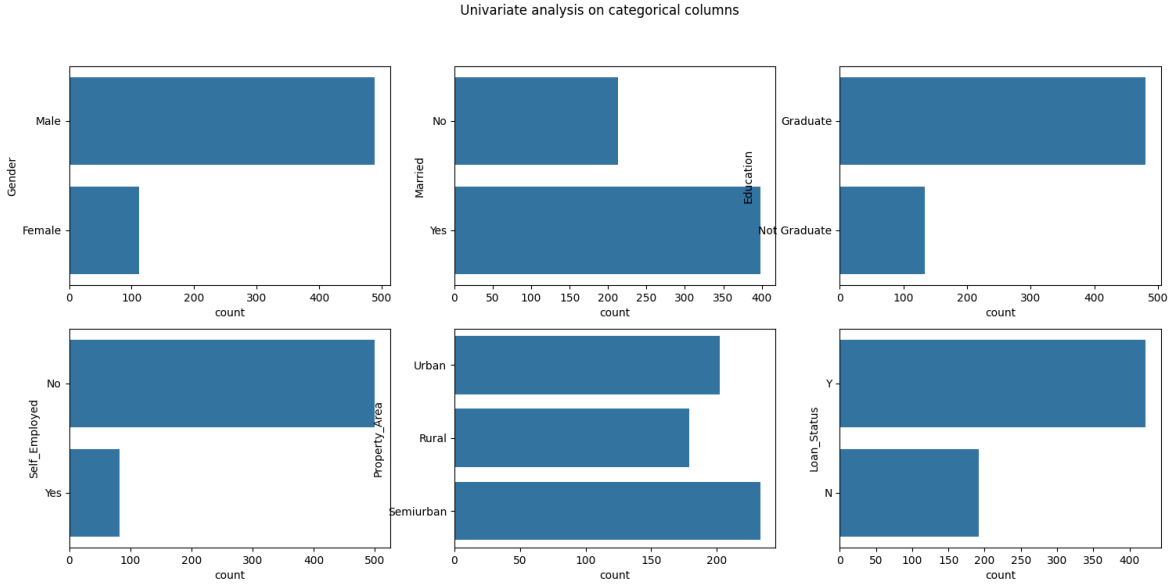
In [50]:
```
data[descrete_categorical].describe(include='object')
```

Out[50]:

	Gender	Married	Education	Self_Employed	Property_Area	Loan_Status
count	601	611	614	582	614	614
unique	2	2	2	2	3	2
top	Male	Yes	Graduate	No	Semiurban	Y
freq	489	398	480	500	233	422

For Descrete variable apply count plot

In [52]:
```
plt.rcParams['figure.figsize']=(18,8)

plt.subplot(2,3,1)
sns.countplot(data['Gender'])

plt.subplot(2,3,2)
sns.countplot(data['Married'])

plt.subplot(2,3,3)
sns.countplot(data['Education'])

plt.subplot(2,3,4)
sns.countplot(data['Self_Employed'])

plt.subplot(2,3,5)
sns.countplot(data['Property_Area'])

plt.subplot(2,3,6)
sns.countplot(data['Loan_Status'])

plt.suptitle(' Univariate analysis on categorical columns')
plt.show()
```



For the descrete variables applied the crosstab between the 'Loan_Status' and other descrete categorical variables

In [54]:
```
print("Impact of Marriage on Loan_Status")
print(pd.crosstab(data['Loan_Status'],data['Married']))
print('\n')

print("Impact of Dependents on Loan_Status")
print(pd.crosstab(data['Loan_Status'],data['Dependents']))
print('\n')

print("Impact of Education on Loan_Status")
print(pd.crosstab(data['Loan_Status'],data['Education']))
print('\n')

print("Impact of Employment on Loan_Status")
print(pd.crosstab(data['Loan_Status'],data['Self_Employed']))
print('\n')

print("Impact of Property on Loan_Status")
print(pd.crosstab(data['Loan_Status'],data['Property_Area']))
print('\n')
```

```
Impact of Marriage on Loan_Status
Married       No  Yes
Loan_Status
N             79  113
Y            134  285


Impact of Dependents on Loan_Status
Dependents     0   1   2  3+
Loan_Status
N            107  36  25  18
Y            238  66  76  33


Impact of Education on Loan_Status
Education    Graduate  Not Graduate
Loan_Status
N                 140            52
Y                 340            82


Impact of Employment on Loan_Status
Self_Employed   No  Yes
Loan_Status
N              157   26
Y              343   56


Impact of Property on Loan_Status
Property_Area  Rural  Semiurban  Urban
Loan_Status
N                 69         54     69
Y                110        179    133
```

Checked for Missing Values

In [56]:
```
data.isnull().sum()
```

Out[56]:
```
Gender              13
Married              3
Dependents          15
Education            0
Self_Employed       32
LoanAmount          22
Loan_Amount_Term    14
Credit_History      50
Property_Area        0
Loan_Status          0
Income               0
dtype: int64
```

Checked for Missing Skewness

In [58]:
```
data[continous].skew()
```

Out[58]:
```
Income        5.633449
LoanAmount    2.677552
dtype: float64
```

Those skewness values indicate that both the Income and LoanAmount distributions are highly right-skewed (positively skewed)

What skewness means numerically: Skewness = 0 → perfectly symmetrical (e.g., bell-shaped)

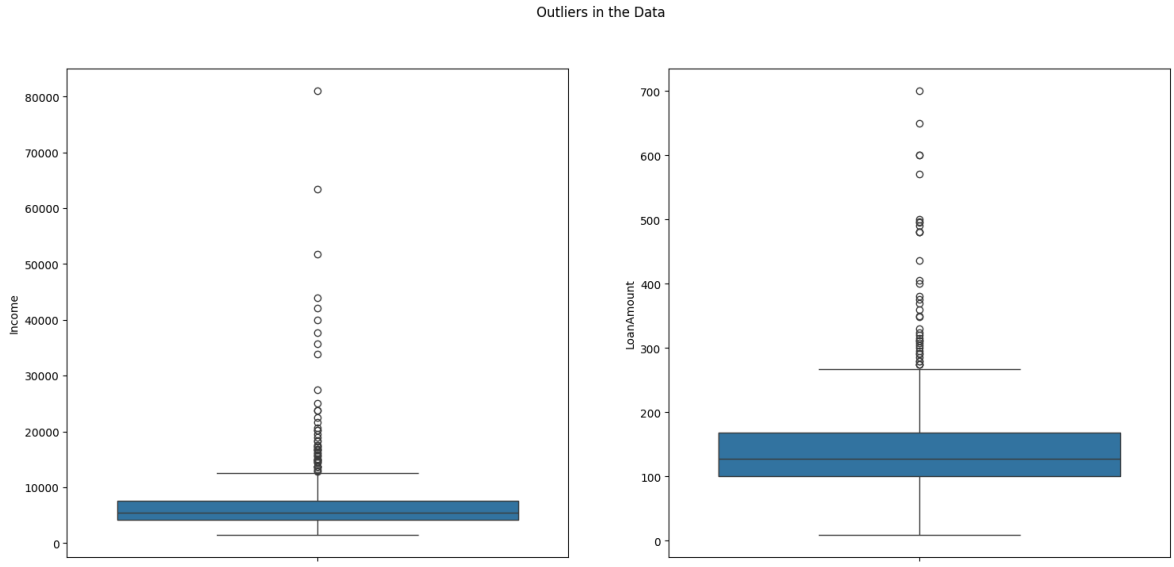
Skewness > 1 → highly positively skewed

Skewness between 0.5 and 1 → moderately skewed

Skewness < -1 → highly negatively skewed

Checked for Outliers

In [62]:
```
plt.subplot(1,2,1)
sns.boxplot(data['Income'])

plt.subplot(1,2,2)
sns.boxplot(data['LoanAmount'])

plt.suptitle('Outliers in the Data')
plt.show()
```



Income & LoanAmount Both Having Outliers ---> but we cant consider then as outliers as some people can have the high income or can apply high loan amount

3) Data Preparation

1)- Data Cleaning --> 1) Wrong Data 2) Missing Values 3) Wrong data types 4) Duplicates 5) Outliers

2)- Data Wrangling --> 1) Transformation 2) Scaling 3) Encoding

1) Data Cleaning

Wrong Data Treatment

In [68]:
```
data['Dependents'] = data['Dependents'].replace({'3+':3})
```

In [69]:
```
data['Dependents'].unique()
```

Out[69]:
```
array(['0', '1', '2', 3, nan], dtype=object)
```

Missing values Treatment

In [71]:
```
data['Dependents'] = data['Dependents'].fillna(0)

data['Gender'] = data['Gender'].fillna(data['Gender'].mode()[0])

data['Married'] = data['Married'].fillna(data['Married'].mode()[0])

data['Self_Employed'] = data['Self_Employed'].fillna(data['Self_Employed'].mode(

data = data.dropna(subset = ['Income', 'LoanAmount', 'Loan_Amount_Term', 'Credit
# As Loan will be depending on these variables so we can't replace these variabl
```

Data Types Conversions

In [73]:
```
data['Dependents'] = data['Dependents'].astype('int')

data['Loan_Amount_Term'] = data['Loan_Amount_Term'].astype('int')
```

Duplicates Treatments

In [75]:
```
data.duplicated().sum()
```

Out[75]:
```
0
```

Outliers Treatments

In [77]:
```
# Outliers should be retrained(Really some people will have high income)
```

2) Data Wrangling

Encoding

In [80]:
```
data.columns
```

Out[80]:
```
Index(['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed',
       'LoanAmount', 'Loan_Amount_Term', 'Credit_History', 'Property_Area',
       'Loan_Status', 'Income'],
      dtype='object')
```

In [81]:
```
data['Gender'] = data['Gender'].replace({'Male':1, 'Female':0})
data['Married'] = data['Married'].replace({'Yes':1, 'No':0})
data['Education'] = data['Education'].replace({'Graduate':1, 'Not Graduate':0})
data['Self_Employed'] = data['Self_Employed'].replace({'Yes':1, 'No':0})
data['Credit_History'] = data['Credit_History'].replace({'good':1, 'bad':0})
data['Property_Area'] = data['Property_Area'].replace({'Rural':0, 'Semiurban':1,
data['Loan_Status'] = data['Loan_Status'].replace({'Y':1, 'N':0})
```

Transformations

for the right skewed---> Go for Transformation

In [83]:
```
from scipy.stats import boxcox
data['Income'], a = boxcox(data['Income'])
data['LoanAmount'], c = boxcox(data['LoanAmount'])
```

Checked the skewness for Income & LoanAmount

In [85]:
```
data['Income'].skew()
```

Out[85]:
```
-0.02776908801128399
```

In [86]:
```
data['LoanAmount'].skew()
```

Out[86]:
```
0.03828926265578848
```

Scaling

I have checked Income & LoanAmount --> Both have same scale ..so no need to apply scaling on that

Also no need to do scaling on count variables

X & y

In [90]:
```
X = data.drop('Loan_Status', axis=1)
y = data['Loan_Status']
```

In [91]:
```
y = data['Loan_Status']
```

In [92]:
```
y.value_counts()
```

Out[92]:
```
Loan_Status
1    366
0    163
Name: count, dtype: int64
```

train-test split

In [94]:
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_
```

4) Modeling & Evaluation

In [96]:
```
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.ensemble import GradientBoostingClassifier
from xgboost import XGBClassifier

from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score
from sklearn.model_selection import cross_val_score
```

1) Logistic Regression

In [98]:
```
# Modeling
log_model = LogisticRegression()
log_model.fit(X_train, y_train)
```

Out[98]:
```
LogisticRegression()
```

In [99]:
```
# Evaluation
ypred_train = log_model.predict(X_train)
ypred_test = log_model.predict(X_test)
```

In [100…
```
print("Train Accuracy: ", accuracy_score(y_train, ypred_train) )
print("CV Score: ", cross_val_score(log_model, X_train, y_train, cv=5, scoring='
print("Test Accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train Accuracy:  0.8226950354609929
CV Score:  0.8227731092436976
Test Accuracy:  0.7830188679245284
```

2) KNN

- HPT
- Model & Evaluation

In [103…
```
# Hyper parameter Tuning

estimator = KNeighborsClassifier()
param_grid = {'n_neighbors': list(range(1,50))}

knn_grid = GridSearchCV(estimator, param_grid, scoring='accuracy', cv=5)
knn_grid.fit(X_train, y_train)

knn_grid.best_estimator_
```

Out[103…
```
KNeighborsClassifier(n_neighbors=12)
```

In [104…
```
# Modeling
knn_model = KNeighborsClassifier(n_neighbors=12)
knn_model.fit(X_train, y_train)
```

Out[104…
```
KNeighborsClassifier(n_neighbors=12)
```

In [105…
```
# Evaluation
ypred_train = knn_model.predict(X_train)
ypred_test = knn_model.predict(X_test)
```

In [106…
```
print("Train Accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(knn_model, X_train, y_train, scoring='accura
print("Test Accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train Accuracy:  0.7683215130023641
CV Score:  0.7375630252100841
Test Accuracy:  0.7358490566037735
```

3) SVM (Support Vector Machine)

- HPT
- Modeling & Evaluation

In [109…
```
# Hyperparameter Tuning

estimator = SVC()
param_grid = {'C':[0.01,0.1, 1], 'kernel':['linear', 'rbf','sigmoid','poly']}

svm_grid = GridSearchCV(estimator, param_grid, scoring = 'accuracy', cv=5)
svm_grid.fit(X_train, y_train)

svm_grid.best_estimator_
```

Out[109…
```
SVC(C=0.1, kernel='linear')
```

In [110…
```
# Modeling
svm_model = SVC(C=0.1, kernel='linear')
svm_model.fit(X_train, y_train)
```

Out[110…
```
SVC(C=0.1, kernel='linear')
```

In [111…
```
# Evaluation
ypred_train = svm_model.predict(X_train)
ypred_test = svm_model.predict(X_test)

print("Train accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(svm_model, X_train, y_train, scoring='accura
print("Test accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train accuracy:  0.8226950354609929
CV Score:  0.8227731092436976
Test accuracy:  0.7830188679245284
```

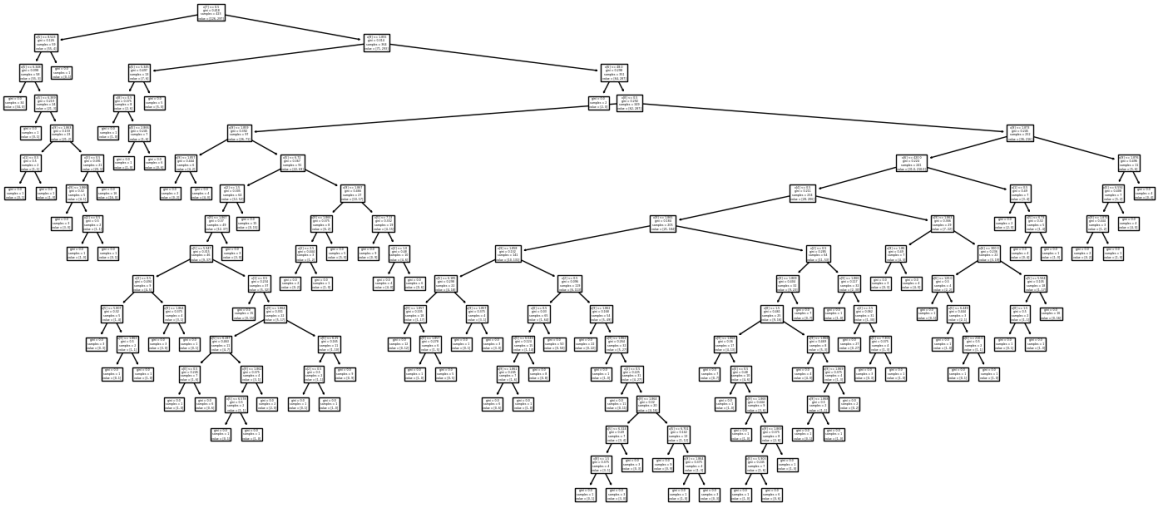
4) Decision Tree Classifier

- HPT
- Identified important feature by using feature selection (ensemble method)
- Modeling & Evaluation

In [114…
```
model = DecisionTreeClassifier(random_state = True)
model.fit(X_train, y_train)
```

Out[114…
```
DecisionTreeClassifier(random_state=True)
```

In [115…
```
from sklearn.tree import plot_tree
plot_tree(model)
plt.show()
```



In [116…
```
# Hyperparameter Tuning

estimator = DecisionTreeClassifier(random_state=True)
param_grid = {'criterion': ['gini', 'entropy'],
              'max_depth': list(range(1,16))}

dt_grid = GridSearchCV(estimator, param_grid, scoring = 'accuracy', cv=5 )
dt_grid.fit(X_train, y_train)

dt = dt_grid.best_estimator_
dt.feature_importances_
```

Out[116…
```
array([0., 0., 0., 0., 0., 0., 0., 1., 0., 0.])
```

In [117…
```
# Important Parameters

feats_dt = pd.DataFrame(data = dt.feature_importances_,
                        index = X.columns,
                        columns=['Importance'] )

important_features_dt = feats_dt[feats_dt['Importance']>0].index.tolist()

important_features_dt
```

Out[117…
```
['Credit_History']
```

Creating Decision Tree Model with important parameters and important features

In [119…
```
# Selecting train & test data

X_train_dt = X_train[important_features_dt]
X_test_dt = X_test[important_features_dt]
```

In [120…
```
# Modeling

dt.fit(X_train_dt, y_train)
```

Out[120…
```
DecisionTreeClassifier(max_depth=1, random_state=True)
```

In [121…
```
# Evaluation

ypred_train = dt.predict(X_train_dt)
ypred_test = dt.predict(X_test_dt)
```

In [122…
```
print("Train accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(dt, X_train, y_train, scoring='accuracy', cv
print("Test accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train accuracy:  0.8226950354609929
CV Score:  0.8227731092436976
Test accuracy:  0.7830188679245284
```

5) Random Forest Classifier

- HPT
- Identified important feature by using feature selection (ensemble method)
- Modeling & Evaluation

In [125…
```
# Hyperparameter Tuning

estimator = RandomForestClassifier(random_state=True)
param_grid = {'n_estimators': list(range(1, 51))}

rf_grid = GridSearchCV(estimator, param_grid, scoring='accuracy', cv=5)
rf_grid.fit(X_train, y_train)
rf = rf_grid.best_estimator_
rf
```

Out[125…
```
RandomForestClassifier(n_estimators=21, random_state=True)
```

In [126…
```
# Important features

feats_ab = pd.DataFrame(data = rf.feature_importances_,
                        index = X.columns,
                        columns = ['Importance'])

important_features_rf = feats_ab[feats_ab['Importance']>0].index.tolist()
important_features_rf
```

Out[126…
```
['Gender',
 'Married',
 'Dependents',
 'Education',
 'Self_Employed',
 'LoanAmount',
 'Loan_Amount_Term',
 'Credit_History',
 'Property_Area',
 'Income']
```

In [127…
```
# Selecting train & test data

X_train_rf = X_train[important_features_rf]
X_test_rf = X_test[important_features_rf]
```

In [128…
```
# Modeling

rf.fit(X_train_rf, y_train)
```

Out[128…
RandomForestClassifier

```
RandomForestClassifier(n_estimators=21, random_state=True)
```

In [129…
```
# Evaluation

ypred_train = rf.predict(X_train_rf)
ypred_test = rf.predict(X_test_rf)
```

In [130…
```
print("Train accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(rf, X_train, y_train, scoring='accuracy', cv
print("Test accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train accuracy:  0.9952718676122931
CV Score:  0.8203641456582634
Test accuracy:  0.7735849056603774
```

6) AdaBoost Classifier

- HPT
- Identified important feature by using feature selection (ensemble method)
- Modeling & Evaluation

In [133…
```
# Hyperparameter Tuning

estimator = AdaBoostClassifier(random_state=True)
param_grid = {'n_estimators': list(range(1, 51))}

ab_grid = GridSearchCV(estimator, param_grid, scoring='accuracy', cv=5)
ab_grid.fit(X_train, y_train)
ab = ab_grid.best_estimator_
ab
```

Out[133…
AdaBoostClassifier

```
AdaBoostClassifier(n_estimators=3, random_state=True)
```

In [134…
```
# Important features

feats_ab = pd.DataFrame(data = ab.feature_importances_,
                        index = X.columns,
                        columns = ['Importance'])

important_features_ab = feats_ab[feats_ab['Importance']>0].index.tolist()
```

```
important_features_ab
```

Out[134…
```
['Credit_History', 'Property_Area', 'Income']
```

In [135…
```
# Selecting train & test data

X_train_ab = X_train[important_features_ab]
X_test_ab = X_test[important_features_ab]
```

In [136…
```
# Modeling

ab.fit(X_train_ab, y_train)
```

Out[136…
```
AdaBoostClassifier(n_estimators=3, random_state=True)
```

In [137…
```
# Evaluation

ypred_train = ab.predict(X_train_ab)
ypred_test = ab.predict(X_test_ab)
```

In [138…
```
print("Train accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(ab, X_train, y_train, scoring='accuracy', cv
print("Test accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train accuracy:  0.83451536643026
CV Score:  0.8322408963585435
Test accuracy:  0.7641509433962265
```

7) Gradient Boost Classifier

- HPT
- Identified important feature by using feature selection (ensemble method)
- Modeling & Evaluation

In [141…
```
# Hyperparameter Tuning

estimator = GradientBoostingClassifier(random_state=True)
param_grid = {'n_estimators': list(range(1, 10)),
              'learning_rate': [0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 0.8, 0.9, 1.0]}

gb_grid = GridSearchCV(estimator, param_grid, scoring='accuracy', cv=5)
gb_grid.fit(X_train, y_train)
gb = gb_grid.best_estimator_
gb
```

Out[141…
```
GradientBoostingClassifier(learning_rate=0.3, n_estimators=3, random_st
ate=True)
```

In [142…
```
# Important features

feats_gb = pd.DataFrame(data = gb.feature_importances_,
                        index = X.columns,
                        columns = ['Importance'])

important_features_gb = feats_gb[feats_gb['Importance']>0].index.tolist()
important_features_gb
```

Out[142…
```
['Married',
 'LoanAmount',
 'Loan_Amount_Term',
 'Credit_History',
 'Property_Area',
 'Income']
```

In [143…
```
# Selecting train & test data

X_train_gb = X_train[important_features_gb]
X_test_gb = X_test[important_features_gb]
```

In [144…
```
# Modeling

gb.fit(X_train_gb, y_train)
```

Out[144…
```
GradientBoostingClassifier
GradientBoostingClassifier(learning_rate=0.3, n_estimators=3, random_st
ate=True)
```

In [145…
```
# Evaluation

ypred_train = gb.predict(X_train_gb)
ypred_test = gb.predict(X_test_gb)
```

In [146…
```
print("Train accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(gb, X_train, y_train, scoring='accuracy', cv
print("Test accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train accuracy:  0.8321513002364066
CV Score:  0.8227450980392158
Test accuracy:  0.7735849056603774
```

8) XG Boost Classifier

- HPT
- Identified important feature by using feature selection (ensemble method)
- Modeling & Evaluation

In [149…
```
# Hyperparameter Tuning

estimator = XGBClassifier()
param_grid = {'n_estimators': [10, 20, 40, 100],
```

```
            'max_depth': [3, 4, 5],
            'gamma': [0, 0.15, 0.3, 0.5, 1]}

xgb_grid = GridSearchCV(estimator, param_grid, scoring='accuracy', cv=5)
xgb_grid.fit(X_train, y_train)
xgb = xgb_grid.best_estimator_
xgb
```

Out[149…

XGBClassifier

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rou
nds=None,
              enable_categorical=False, eval_metric=None, feature_ty
pes=None,
              feature_weights=None, gamma=0.3, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=No
```

In [150…

```
# Important features

feats_xgb = pd.DataFrame(data = xgb.feature_importances_,
                         index = X.columns,
                         columns = ['Importance'])

important_features_xgb = feats_xgb[feats_xgb['Importance']>0].index.tolist()
important_features_xgb
```

Out[150…

```
['Dependents',
 'Self_Employed',
 'LoanAmount',
 'Loan_Amount_Term',
 'Credit_History',
 'Property_Area',
 'Income']
```

In [151…

```
# Selecting train & test data

X_train_xgb = X_train[important_features_xgb]
X_test_xgb = X_test[important_features_xgb]
```

In [152…

```
# Modeling

xgb.fit(X_train_xgb, y_train)
```

Out[152...

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rou
nds=None,
              enable_categorical=False, eval_metric=None, feature_ty
pes=None,
              feature_weights=None, gamma=0.3, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=No
```

In [153...

```
# Evaluation

ypred_train = xgb.predict(X_train_xgb)
ypred_test = xgb.predict(X_test_xgb)
```

In [154...

```
print("Train accuracy: ", accuracy_score(y_train, ypred_train))
print("CV Score: ", cross_val_score(xgb, X_train, y_train, scoring='accuracy', c
print("Test accuracy: ", accuracy_score(y_test, ypred_test))
```

```
Train accuracy:  0.8321513002364066
CV Score:  0.8251260504201682
Test accuracy:  0.7830188679245284
```

Save the best model

In [156...

```
from joblib import dump
dump(dt,'loan.joblib')
```

Out[156...

```
['loan.joblib']
```

Predict on new data

In [158...

```
input_data = {
    "Loan_ID": "LP002991",
    "Gender": "Male",
    "Married": "No",
    "Dependents": 1,
    "Education": "Graduate",
    "Self_Employed": "yes",
    "ApplicantIncome": 1000,
    "CoapplicantIncome": 0,
    "LoanAmount": 100,
    "Loan_Amount_Term": 240,
    "Credit_History": "bad",
    "Property_Area": "Urban"
}
```

In [159...

```
data  = pd.DataFrame(input_data, index=[0])
data
```

Out[159...

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome
0	LP002991	Male	No	1	Graduate	yes	100(

Apply Data Preprocessing on Unknown Data

In [161...

```
# dropped unimportant column
data.drop(columns=["Loan_ID"], inplace=True)

# Created new column
data["Income"] = data["ApplicantIncome"] + data["CoapplicantIncome"]
data.drop(columns=['ApplicantIncome','CoapplicantIncome'], inplace=True)

# Handeled missing values
data['Dependents'] = data['Dependents'].fillna(0)
data['Gender'] = data['Gender'].fillna(data['Gender'].mode()[0])
data['Married'] = data['Married'].fillna(data['Married'].mode()[0])
data['Self_Employed'] = data['Self_Employed'].fillna(data['Self_Employed'].mode(
data = data.dropna(subset=["Income",'LoanAmount','Loan_Amount_Term','Credit_Hist

# Treated wrong data types
data['Dependents'] = data['Dependents'].astype('int')
data['Loan_Amount_Term'] = data['Loan_Amount_Term'].astype('int')

# Encoding
data['Gender'] = data['Gender'].replace({'Male':1,'Female':0})
data['Married'] = data['Married'].replace({'Yes':1,'No':0})
data['Education'] = data['Education'].replace({'Graduate':1,'Not Graduate':0})
data['Self_Employed'] = data['Self_Employed'].replace({'Yes':1,'No':0})
data['Property_Area'] = data['Property_Area'].replace({'Rural':0,'Semiurban':1,'
data['Credit_History'] = data['Credit_History'].replace({'good':1,'bad':0})

X_new = data
```

In [162...

```
X_new
```

Out[162...

	Gender	Married	Dependents	Education	Self_Employed	LoanAmount	Loan_Amour
0	1	0	1	1	yes	100	

Select important features of your best model

In [164...

```
X_new = X_new[['Credit_History']]
```

Apply and predict using your best model

In [166...

```
dt.predict(X_new)
```

Out[166...

```
array([0], dtype=int64)
```

i.e Not Eligible